Seat No.: _____ Enrolment No._____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER–III (NEW) EXAMINATION – WINTER 2020

Subject Code:3130702 **Date:10/03/2021**

Subject Name:Data Structures

Time:10:30 AM TO 12:30 PM **Total Marks:56**

Instructions:

1. **Attempt any FOUR questions out of EIGHT questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**

			Marks
Q.1	**(a)**	Compare array and linked list.	**03**
	(b)	Compare primitive and non primitive data types. datastructures	**04**
	(c)	Write an algorithm to perform insert and delete operations on simple queue.	**07**
Q.2	**(a)**	Search the number 50 from the given data using binary search technique. Illustrate the searching process. 10, 14, 20, 39, 41, 45, 49, 50, 60	**03**
	(b)	Apply merge sort algorithm to the following elements. 20, 10, 5, 15, 25, 30, 50, 35	**04**
	(c)	Write a 'C' program for bubble sort.	**07**
Q.3	**(a)**	What is stack? Why do we use multiple stacks?	**03**
	(b)	Convert the following infix expressions to their prefix and postfix equivalents. 1. A*B+C/D 2. (A*B)+(C/D)-(D+E)	**04**
	(c)	What is priority queue? Discuss its applications and implementation details.	**07**
Q.4	**(a)**	Evaluate the following postfix expression using stack. 53+62/*35*+	**03**
	(b)	Design an algorithm to perform insert operation in circular queue.	**04**
	(c)	Design an algorithm to merge two linked list.	**07**
Q.5	**(a)**	Define: 1. Acyclic graph 2. Leaf node 3. Complete binary tree	**03**
	(b)	For following expressions, construct the corresponding binary tree. 1. A+B/C*D-E 2. ((A+B)-(C*D))%((E^F)/(G-H))	**04**
	(c)	How are graphs represented inside a computer's memory? Which method do you prefer and why?	**07**
Q.6	**(a)**	Define: 1. Connected graph 2. Threaded tree 3. Degree of node	**03**
	(b)	Differentiate between depth first search and breadth first search.	**04**
	(c)	Design an algorithm to insert a given value in the binary search tree.	**07**
Q.7	**(a)**	Explain basic file operations.	**03**
	(b)	List out applications of hashing.	**04**
	(c)	What is file organization? Briefly summarize different file organizations.	**07**

Q.8	**(a)**	Give a brief note on indexing.	**03**
	(b)	Build a chained hash table of 10 memory locations. Insert the keys 131, 3, 4, 21, 61, 24, 7, 97, 8, 9 in hash table using chaining. Use h(k) = k mod m. (m=10)	**04**
	(c)	Consider the hash table of size 10. Using quadratic probing, insert the keys 72, 27, 36, 24, 63, 81, and 101 into hash table. Take c_1=1 and c_2=3.	**07**

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